Final 2 ns water analysis

Completion

All 168 canonical trajectories reached 2 ns NVT (336 ns total), with 1 ns NPT each. The final GPU job completed October 25 at 20:45:20 UTC; full-study CPU job 59930836 completed at 23:30:41 UTC. All 2,688 segment records and exact final restarts passed the current audit. Historical LES trajectories are excluded.

Scientific correction

The historical LES wrapper duplicated the cell-strain contribution already implemented natively. This affected both NPT pressure/density and NVT stress. All 84 LES trajectories were rerun with the validated native implementation. The old 2.4-fold viscosity split and persistent-mode interpretation must be removed. Molecular model weights and RMSE data are unchanged.

Corrected 295 K result

G1–G6 OFF: eta = 0.977 ± 0.145 mPa s; LES: 1.180 ± 0.094 mPa s. Their ratio of means is 1.207, a descriptive comparison. Error bars are sample SD over three fitted-seed means, each averaging three MD replicas. Raw D is 1.766 ± 0.191 (OFF) and 1.453 ± 0.131 (LES) nm²/ns; corrected D is 2.262 ± 0.259 and 1.866 ± 0.163 nm²/ns.

Temperature-dependent agreement

At 275 K, broad LES viscosity is 60.2% above experiment, versus 14.7% below for the baseline. At 336 K, OFF corrected-diffusion error rises from 8.0% under narrow coverage to 23.8% under broad coverage. All 24 state errors are tabulated. These comparisons concern experiment, not fidelity to the quantum reference.

Convergence evidence

Median per-trajectory absolute symmetric changes from 1.5 to 2 ns are 0.69% for raw diffusion and 2.10% for viscosity. Between disjoint first/second nanoseconds, median viscosity change is 9.08%; 18/168 exceed 20%. Comparing viscosity windows, 12/168 lanes and 0/24 means shift by >10% at 10–20 ps, but 4/24 means do at 30–50 ps. These diagnostics support a qualified report, not blanket convergence.

Plot sequence

Figure 1: RDF above density, OFF left and LES right. Figure 2: diffusion above viscosity, same columns. Figure A1: molecular errors. The archived final-supplement code supplies sizes, margins, fonts, colors and markers. Diffusion is now corrected in the main figure; the raw variant and all new diagnostics are supplementary.

Methods, uncertainty and limits

Estimators

Native 0.5 fs oxygen velocity and five traceless stress records are used without correlation-grid decimation. Duplicate restart-boundary records are removed. Diffusion and viscosity remain the prespecified 5–10 ps running-integral mean. MSD fits use 10–600 ps. The additional stress diagnostic extends the plotted lag to 50 ps and independently reproduces every primary viscosity and 100 ps block-prefix estimate.

Sampling units

All 56 model/seed/temperature states have three MD replicas. Adapted conditions use training seeds 17, 29, 43. Seed-level sample SD follows replica averaging; baselines use MD-replica sample SD. No time origin, stress channel or correlated prefix is counted as an independent model fit. Disjoint windows are labeled disjoint, not certified independent.

Finite size

Each trajectory uses $D_{YH} = D_L + 2.837297\, k_B T/(6 \pi \eta L)$, with its own temperature, volume and viscosity. Aggregation preserves raw-D/correction covariance exactly. This is an estimated hydrodynamic correction for 64-water cells, not a direct thermodynamic-limit extrapolation. Viscosity-size dependence remains unresolved.

Block sensitivity

Separate 100/200/500 ps blocks are demeaned and correlated separately. Their median absolute symmetric shifts from full-2-ns viscosity are 15.84%, 7.18% and 2.19%. Short-block cumulative averages can therefore appear stable while retaining finite-block bias. They are not substituted for the whole-trajectory estimator.

Remaining limitations

Missing controls include larger cells, matched periodic quantum labels, nuclear quantum effects, alternative adaptation algorithms and LES-readout ablation. G1 lacks exact source-method metadata. Its validation forces also have a 0.897 meV/Å rotational component incompatible with an isolated invariant energy. Coverage has training confounds; experiment cannot establish reference-level fidelity.

Reading and reproducibility

Vector PDFs, PNG previews, lane/seed/state CSVs, exact module lists, archived plotting code, input hashes and validation records are included. Raw MD arrays remain in the frozen campaign. The 24-page appendix shows every model-temperature state. The OpenReview PDF endpoint returned HTTP 403; the supplied review labels and verified archived supplement define the mapping.